Few-Shot Domain Adaptation Transformer for Hyperspectral Image Change Detection

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***Abstract*—Deep learning has shown strong performance in hyperspectral images change detection (HSIs-CD), but it typically requires a large amount of labeled data. Acquiring such labeled data is often expensive, time consuming, and infeasible in many application scenarios. Few-shot learning (FSL) addresses this limitation by training a model on a large number of training tasks and then applying it to test tasks with only a few labeled samples. However, in HSIs-CD, variations in imaging conditions lead to significant distribution gaps across datasets. Consequently, FSL models trained on these training tasks typically generalize poorly to test tasks and may fail to extract meaningful features. To address this, we propose FSDA-Former, a few-shot domain adaptation Transformer for HSIs-CD. First, it employs a generative pretraining strategy based on a masked autoencoder (FSDA-MAE) to learn domain-agnostic representations and improve feature extraction across diverse tasks. Second, an FSDA-Transformer layer (FSDA-TL) is introduced to refine the extracted features in a metalearning manner, thereby learning class-discriminative representations and constructing a task-specific metric space. Third, a two-stage domain adaptation (FSDA-TDA) strategy is developed to further adapt FSDA-former to the feature domain of the test task. FSDA-TDA further enhances the generalization ability of FSDA-former under distribution shifts between training and test tasks. Experimental results on three real-world HSIs-CD datasets demonstrate that FSDA-former consistently outperforms existing few-shot baselines. It achieves up to 3.60% improvement in kappa and 1.58% in overall accuracy (OA). These results demonstrate the effectiveness and superiority of FSDA-former in HSIs-CD, even when only a few labeled examples are available.**

***Index Terms*—Change detection (CD), domain adaptation (DA), few-shot learning (FSL), generative pretrained Transformer, hyperspectral images (HSIs).**

Received 15 April 2026; accepted 29 May 2026. Date of publication 8 June 2026; date of current version 16 June 2026. This work was supported in part by the National Key Laboratory on Electromagnetic Environmental Effects and Electro-Optical Engineering under Grant KY3240020001 and in part by the Aerospace Science and Technology Innovation Development Fund under Grant ZY0110020009. *(Corresponding authors: Jianjun Sha; Kai Qin.)*

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Digital Object Identifier 10.1109/TGRS.2026.3700816

I. Introduction

HYPERSPECTRAL images (HSIs) capture rich spectral information across hundreds of narrow bands, ranging from ultraviolet to midinfrared. Since each material exhibits a unique spectral signature, HSIs enable fine-grained analysis of the Earth's surface [1], [2], [3]. HSIs change detection (HSIs-CD) assesses land changes by comparing multitemporal HSIs of the same area, proving valuable for agriculture, environmental monitoring, land use planning, urban development, and military applications [4], [5].

Traditional HSIs-CD methods are typically categorized into algebraic and machine learning-based approaches. Among the algebraic methods, change vector analysis (CVA) [6] has been widely used in unsupervised settings, leading to various enhancements like compressed CVA [7], which projects image features into a 2-D space, and sequential spectral CVA [8], developed to better detect small objects. Common machine learning methods in HSIs-CD include principal component analysis (PCA) [9], support vector machines (SVMs) [10], and K-means clustering [11], which are computationally efficient but limited in feature extraction, often resulting in lower accuracy.

With the rise of deep learning, HSIs-CD has witnessed significant improvements in feature extraction and representation. Deep learning-based approaches in HSIs-CD generally include unsupervised, semisupervised, and supervised learning. Unsupervised learning methods eliminate the need for labeled data by leveraging autoencoders and pretrained networks on auxiliary tasks to extract meaningful features for HSIs-CD [12], [13]. For instance, Lei et al. [14] utilized an autoencoder to extract features from bitemporal HSIs and adopted an adversarial learning strategy to align their representations within a shared feature space. Liu et al. [15] designed a multitask autoencoder that integrates HSIs reconstruction and edge detection to extract features. Leenstra et al. [16] utilized pretext tasks to pretrain the network, then used the pretrained network to extract features from bitemporal HSIs and classified the extracted features using CVA. The second approach involves training networks using pseudolabels generated by traditional unsupervised methods. Shi et al. [17] adopted a self-guided approach to progressively generate pseudolabels for training the network. Seydi and Hasanlou [18] used Otsu and dynamic time wrapping to obtain pseudolabels and then trained a 3-D convolutional neural network (CNN) to extract multitemporal image features. Qu et al. [19] designed a progressive graph-to-pixel joint constraint strategy to generate pseudolabels and improve HSIs-CD performance through domain adaptation

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(DA). Despite promising results, these methods often struggle in scenarios where the same objects exhibit significant spectral variability or when only a single class undergoes change.

Supervised learning leverages a large number of labeled samples to learn complex feature representations. It has achieved high accuracy in HSIs-CD. The most commonly used methods include CNNs, Graph convolutional networks (GCNs), attention mechanisms, and Transformers. CNNs have strong spatial feature extraction capabilities. For example, Seydi et al. [20] combined 3-D, 2-D, and 1-D CNNs to extract multidimensional features from HSIs. Ou et al. [21] used CNNs to extract multiscale features and then performed HSIs-CD through multiscale feature fusion. Unlike CNNs, which operate on regular grid-like data, GCNs handle irregular and non-Euclidean data by aggregating and combining features from neighboring nodes based on the graph structure. GCNs have emerged as a powerful tool for HSIs-CD due to their capability to capture complex spatial–spectral relationships within graph-structured data. For example, Yang et al. [22] used GCNs to extract multilevel differential features from HSIs and fused spectral–spatial augmentation information with difference information to obtain discriminative textural features. Attention mechanisms are primarily used to focus on and extract key information from images. The spectral–spatial attention mechanism is combined to identify the spectral bands and spatial locations in HSIs that contribute significantly to CD [23], [24], [25], [26]. Transformers, equipped with self-attention mechanisms and positional encoding, can effectively process sequential data and learn complex relationships between elements. In HSIs-CD, Transformer-based methods have been shown to effectively model spectral and spatial sequences, enabling better extraction of spectral–spatial features [27], [28], [29]. In addition, some studies have extended Transformers to capture temporal dependencies across multitemporal HSIs, aiming to analyze temporal change patterns more effectively [30]. However, high-accuracy CD typically relies on a large number of labeled samples, which are often expensive and time consuming to obtain, especially for high-dimensional HSIs.

To mitigate the reliance on large-scale annotations, semisupervised learning combines a small amount of labeled data with unsupervised features learned from unlabeled HSIs-CD samples. For example, a network was trained to extract HSI features under the supervision of pseudolabels generated from PCA-derived principal components, effectively guiding the unsupervised learning process. The features were then classified using a limited number of labeled samples [31]. Wang et al. [32] utilized a cycle autoencoder to extract unsupervised features that mitigate temporal differences in the same objects, and further integrated a small number of labeled samples to fine-tune the autoencoder and train a multilayer perceptron for CD. Qu et al. [33] first trained the network using a limited number of samples and then employed a teacher–student self-distillation framework to reduce annotation dependency while maintaining sensitivity to change pixels. In [34], a weak-to-strong consistency was introduced to improve feature extraction by leveraging both labeled and unlabeled data. Wang et al. [35] introduced a lightweight masked autoencoder for HSIs that supports semisupervised HSIs-CD by allowing efficient fine-tuning on downstream tasks using only a limited number of labeled samples.

Foundation models, trained on large-scale multidomain unlabeled datasets, exhibit strong generalization capability, enabling effective feature extraction for unseen CD tasks. SatMAE [36] and SpectralGPT [37] are foundation models for spectral data with a few tens of spectral bands, while HyperSIGMA [38], HyperFree [39], and HyperSL [40] are specifically designed for HSIs with hundreds of spectral bands. Despite their ability to yield highly discriminative features for HSIs-CD with limited training samples, the massive parameter size and high computational demands of foundation models pose significant obstacles to operational deployment, especially in resource-limited settings.

Few-shot learning (FSL) offers an effective solution to the challenge of scarce labeled data by enabling models to generalize from only a few annotated examples. It has shown promising results in computer vision and HSIs classification [41], [42], [43], [44]. In a typical FSL paradigm, models are trained on a series of training tasks, each constructed with a support set and a query set. During testing, the trained model is evaluated on test tasks with unseen classes, where only a few labeled samples are available as a support set. Despite this progress, very few studies have applied FSL to HSIs-CD. Existing methods mainly employ high-resolution remote sensing images as the source of training tasks for HSIs-CD. [45], [46], [47], [48]. However, these approaches are typically trained on paired high-resolution images from the source domain and then applied to extract features from HSIs in the target task. This paradigm inherently constrains the model's ability to extract HSI features. In addition, such methods mainly align HSI features to a fixed metric space constructed in the source domain, which lacks representational flexibility and therefore cannot dynamically adjust distance metrics to effectively compensate for the interference caused by distribution discrepancies.

We adopt a metalearning paradigm in which multiple HSIs-CD datasets are treated as auxiliary tasks, thereby encouraging the model to learn diverse feature representations and construct metric spaces that capture dataset-specific distributions. However, applying this paradigm to few-shot HSIs-CD still involves three major challenges.

1) The auxiliary training datasets and the target test dataset exhibit significant distribution discrepancies arising from sensor variations, imaging conditions, and land-cover differences. The model trained on auxiliary few-shot datasets often fails to extract effective features from the test data due to severe distribution discrepancies.
2) It remains challenging to dynamically produce task-relevant features for constructing adaptive metric spaces. Rather than relying on static mappings, the model must adjust metric criteria according to task semantics.
3) Under the same feature extractor, like foundation models, the features extracted from the auxiliary and test datasets still exhibit distribution discrepancies. As a result, the few-shot model trained on auxiliary tasks has a limited ability to adapt to the test data domain.

In this article, we propose FSDA-Former, a few-shot DA Transformer for HSIs-CD to address the above challenges.

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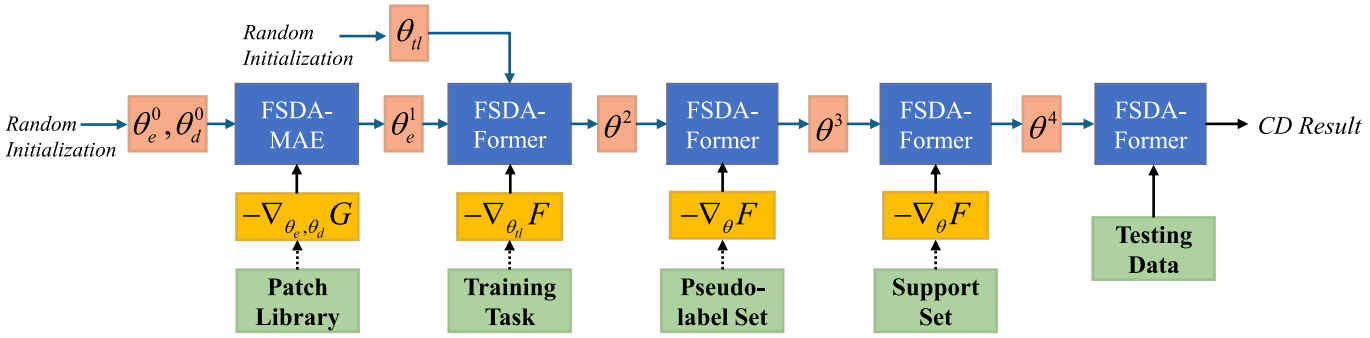


Fig. 1. Flowchart of FSDA-Former: FSDA-MAE is first pretrained using a patch library to update its parameters ($\theta_{\mathbf{e}}^0, \theta_{\mathbf{d}}^0 \rightarrow \theta_{\mathbf{e}}^1, \theta_{\mathbf{d}}^1$). FSDA-Former is trained on the training tasks for FSL to obtain parameters ($\theta^1 = \{\theta_{\mathbf{e}}^1, \theta_{el}^1\} \rightarrow \theta^2$), with $\theta_{\mathbf{e}}^1$ frozen. DA is subsequently performed on the test tasks using pseudolabels and the support set, resulting in parameters ($\theta^2 \rightarrow \theta^3 \rightarrow \theta^4$), which are finally applied for testing.

FSDA-Former consists of three components: the FSDA-masked autoencoder (MAE), the FSDA-Transformer layer (FSDA-TL), and the two-stage DA (FSDA-TDA) strategy. In particular, the FSDA-MAE serves as the backbone of FSDA-Former and is pretrained in a generative manner under an MAE framework, endowing the model with domain-agnostic representation capability for both training and test tasks. Built upon the extracted features, FSDA-TL is introduced to further refine them and efficiently capture class-discriminative information for constructing a task-specific metric space. During the FSL stage, FSDA-Former is trained on the training tasks using metric learning to rapidly learn task-specific class representations. Before testing, FSDA-TDA is performed in two stages to adapt the FSDA-Former to the feature domain of the test task. In the first stage, pseudolabels generated from the test data are used to fine-tune FSDA-Former for initial DA. In the second stage, the support set of the test task (few-shot labeled samples) is further used to refine DA. After these two stages, the FSDA-Former is applied to produce CD results on the test data. The main contributions of this article are as follows.

1) We introduce FSDA-Former, an FSL model for HSIs-CD that alleviates the challenges associated with insufficient labeled training data. FSDA-Former bridges the gap caused by the lack of HSIs-CD datasets as training tasks and effectively captures spectral–spatial priors relevant to HSIs-CD.
2) We propose a lightweight FSDA-MAE that employs smaller patch inputs and a reduced number of Transformer encoder layers, enabling efficient extraction of domain-agnostic features through unsupervised generative pretraining on both training and test task data.
3) A FSDA-TL is proposed, employing a single Transformer encoder layer to extract class-relevant representations from the features obtained from the FSDA-MAE encoder, thereby constructing a task-specific metric space that enables FSDA-Former to rapidly adapt to diverse tasks.
4) To mitigate the feature distribution differences between tasks that remain in FSDA-MAE, we propose FSDA-TDA, a two-stage DA method for the test task. In the first stage, pseudolabels generated via CVA are used to fine-tune the model for initial DA. In the second stage, the test task support set is employed to further perform DA and refine the class prototypes.

II. Proposed Method

The flowchart of proposed FSDA-Former, as shown in Fig. 1, outlines five main processes: domain-agnostic feature learning via FSDA-MAE, FSL within FSDA-Former, DA through pseudolabels and support set optimization, and final testing. Sections II-A–II-D provide a detailed description of FSDA-Former.

A. Problem Statement

In a typical few-shot task, there are two sets: a support set $\mathbf{S} = \{(x_i, y_i)\}_{i=1}^{c\times k}$, with c classes and k labeled examples per class, and a query set, $\mathbf{Q} = \{(x_j)\}_{j=1}^{\|Q\|}$, with $\|Q\|$ unlabeled samples, which shares the same label space. The goal is to learn a classifier on $\mathbf{S}$ that can accurately identify the classes in $\mathbf{Q}$. Due to the limited labeled data per class, training an accurate classifier is challenging. This problem is resolved by using an auxiliary set that is divided into multiple c-ways k-shot training tasks, each with its own support and query sets, allowing metatraining to improve the model's ability to classify based on $\mathbf{S}$. However, for HSIs, significant distributional differences between training and test tasks prevent a model trained on the training tasks via metalearning from effectively learning an accurate classifier from the support set $\mathbf{S}$ of the test tasks.

B. FSDA-MAE for Domain-Agnostic Feature Extraction

We use unsupervised generative pretraining to enable the model to learn domain-agnostic representations across diverse tasks. In particular, a domain-agnostic representation refers to a feature embedding that captures the intrinsic spectral–spatial features of the land cover, remaining invariant to domain-specific discrepancies such as varying sensors, imaging conditions, or geographical locations. However, existing generative pretraining foundation models often have massive parameter sizes, resulting in long inference times and low computational efficiency. To address this issue, we designed FSDA-MAE, illustrated in Fig. 2.

In this work, we define $\{X_i, Y_i\}_{i=1}^n$ as the n training HSIs-CD datasets, and $\{X_t\}$ as the test HSIs-CD dataset. $X_{i/t}$ denoting

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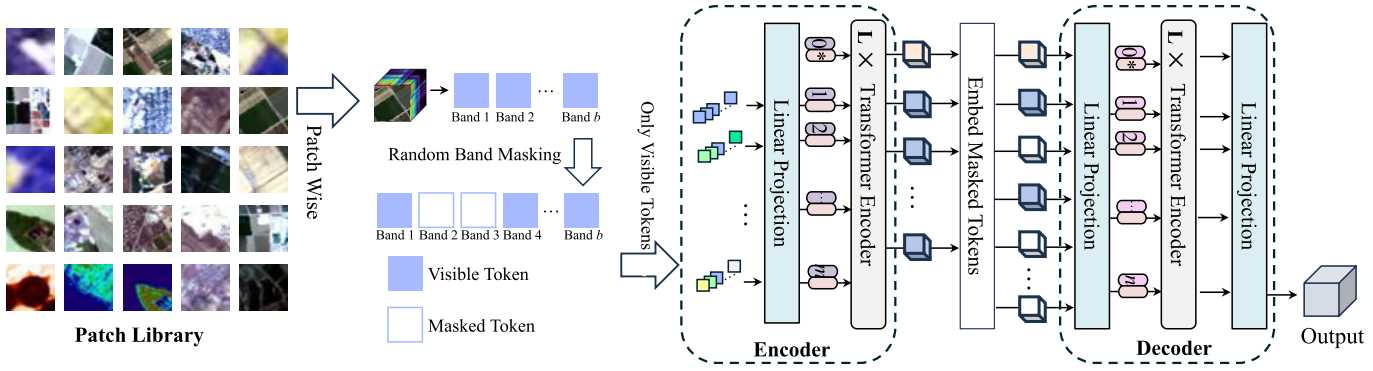


Fig. 2. Flowchart of the FSDA-MAE. The FSDA-MAE is composed of the FSDA-MAE encoder and FSDA-MAE decoder. The FSDA-MAE encoder serves as the backbone of FSDA-Former. The encoder takes as input a spectral sequence composed only of visible tokens, and each token is represented as a 1-D vector obtained by flattening the bandwise information. The blue cubes represent the features of the visible tokens, the pink cubes represent the features of the class tokens, and the white cubes represent the masked tokens.

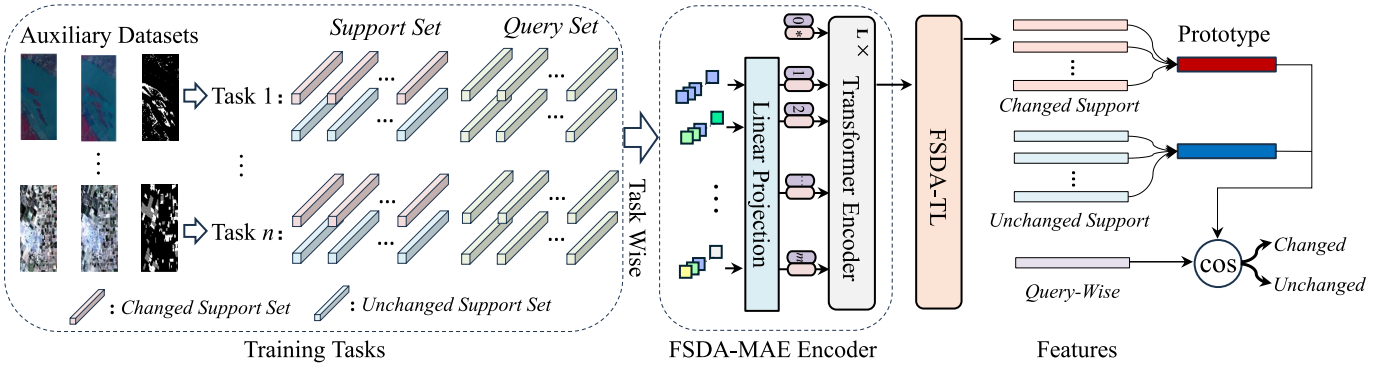


Fig. 3. FSL workflow of FSDA-Former. In each training task, FSDA-Former extracts discriminative representations, with support set features aggregated into class prototypes, while query features are assigned to categories by computing cosine similarity with the corresponding prototypes.

the difference between bitemporal HSIs $\{X^1_{i/t}, X^2_{i/t}\}$, with $X^1_{i/t}$ as the initial time-point image in each set. Y_i serves as the ground truth. To standardize across spectral dimensions, $X_{i/t}$ is resampled to b spectral bands through linear interpolation. We construct a patch library $\mathbf{x}$ by dividing the training sets $\{X_i\}_{i=1}^n$ and test set $\{X_t\}$ into patch with a stride of s. Each patch $x \in \mathbb{R}^{h\times w\times b}$ undergoes random masking, where $p\%$ of spectral bands are masked, creating visible tokens $x_v \in \mathbb{R}^{h\times w\times b\cdot(1-p\%)}$ and masked tokens $x_m \in 0^{h\times w\times(b\cdot p\%)}$. The visible tokens x_v are then reshaped into a spectral sequence, with $b\cdot(1-p\%)$ elements and each element consisting of $h\cdot w$ spatial pixels. A linear prediction (LP) layer reduces the dimension of each element in the sequence from $h\cdot w$ to d. We concatenate a class token x_{class} to the output of the LP layer, and a learnable positional encoding E_{pos} is embedded as shown below

$$z_0 = \left[x_{class}; w_1 \cdot x_v\right)\right] + E_{pos} \tag{1}$$

where w_1 is the trainable weight of the LP layer. Next, z_0 is processed through L-layer Transformer encoder $t_e(\cdot)$, producing the encoded features f

$$f = t_e(z_0). \tag{2}$$

This series of steps defines the FSDA-MAE encoder $E(\cdot)$, yielding the feature $f \in \mathbb{R}^{b\cdot(1-p\%)\times d}$. For further details on the Transformer structure, please refer to [28] and [49].

Then, the masked tokens x_m are reinserted into the encoded features f at their original positions within x, producing $\hat{f} \in \mathbb{R}^{b\times d}$ as the input to the decoder. Similarly, an LP layer transforms the dimension of $\hat{f}$ from d to r, and a learnable positional encoding D_{pos} is embedded, as follows:

$$v_0 = w_2 \cdot \hat{f} + D_{pos} \tag{3}$$

where w_2 represents the LP layer weights. The sequence v_0 is then processed through a L layer Transformer encoder $t_d(\cdot)$, resulting in the sequence v_L. A final LP layer restores these features to their original spatial dimensions $h\cdot w$, reconstructing x as follows:

$$x' = w_3 \cdot t_d(v_0) \tag{4}$$

where w_3 represents the final LP layer weights, and x' is the reconstruction of x. In FSDA-MAE pretraining, the parameters θ_e and θ_d are optimized by minimizing the reconstruction loss L_{pre}

$$L_{pre} = \frac{1}{N}\sum_{i=1}^{N}\left(x_i - x'_i\right)^2 \tag{5}$$

where N is the total number of patches in FSDA-MAE pretraining.

C. FSDA-Former for FSL

Built upon the pretrained FSDA-MAE, we propose FSDA-TL to efficiently capture class-discriminative features within each task. FSDA-Former is composed of a frozen FSDA-MAE encoder and the FSDA-TL, and is trained in a metalearning

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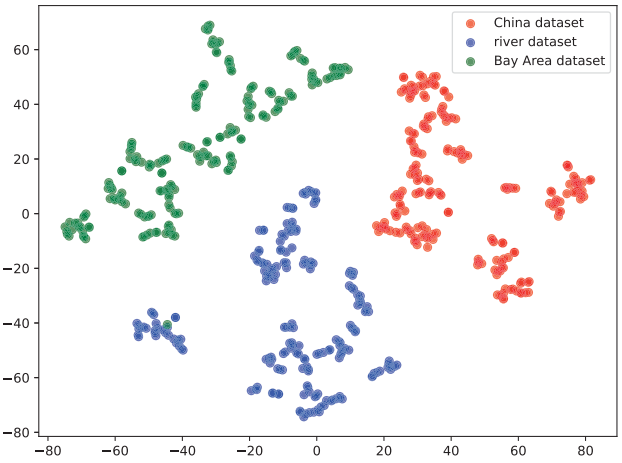


Fig. 4. Distribution of features extracted from the three datasets using the pretrained FSDA-MAE encoder.

manner. The FSDA-MAE encoder is used for feature extraction, while FSDA-TL is designed to learn class-discriminative representations. Its overall architecture is illustrated in Fig. 3. In our FSL approach, we use $\{X_i, Y_i\}_{i=1}^n$ as auxiliary datasets. For each pair X_i, Y_i, we randomly choose $c \cdot k$ pixels, and for each pixel, its surrounding neighborhood is extracted to construct a patch. The patch size is the same as the input patch size of FSDA-MAE, thereby yielding a c-way k-shot training task. In particular, each task ω is first passed through the pretrained FSDA-MAE encoder to extract features $\mathbf{f}_\omega$. These features are further processed by a FSDA-TL $t(\cdot)$ as follows:

$$\mathbf{z} = t\left(\mathbf{f}_\omega = E\left(\omega\right)\right) \tag{6}$$

where $\mathbf{z} = \{\mathbf{z}_\omega^{\mathbf{S}}, \mathbf{z}_\omega^{\mathbf{Q}}\}$ and $t(\cdot)$ is also a standard Transformer encoder. From $\mathbf{z}_\omega^{\mathbf{S}}$, we compute class prototypes A as follows:

$$A_j = \frac{1}{k} \sum_{z_i \in \mathbf{z_j}} z_i, \mathbf{z_j} = \{z_l \mid y_l = j\}, \quad j = 1, \ldots, c, z_l \in \mathbf{z}_\omega^{\mathbf{S}} \tag{7}$$

where c is the number of classes in ω. Each feature in $\mathbf{z}_\omega^{\mathbf{Q}}$ is then assigned to a class based on its similarity to A, as follows:

$$\mathbf{y}' = arg\max_c \frac{\mathbf{z}_\omega^{\mathbf{Q}} \cdot \mathbf{A}}{||\mathbf{z}_\omega^{\mathbf{Q}}|| ||\mathbf{A}||} \tag{8}$$

where $\mathbf{y}'$ is the predicted class of $\mathbf{Q}_\omega$, and $\mathbf{A} = \{A_j\}_{j=1}^c$. The cross-entropy loss between predicted $\mathbf{y}'$ and true labels $\mathbf{y_Q}$ of the query set is used to update the parameters of FSDA-TL, while the FSDA-MAE encoder remains frozen to prevent overfitting during training. We adopt the cross-entropy loss as the optimization objective

$$L_Q = -\sum \mathbf{y_Q} \log\left(\mathbf{y}'\right). \tag{9}$$

The above describes the FSDA-Former training process for a single few-shot task. Finally, we train FSDA-Former across multiple tasks Ω from $\{X_i, Y_i\}_{i=1}^n$, allowing it to quickly learn class-discriminative features using the support set.

D. TDA Strategy

Although the pretrained FSDA-MAE encoder provides domain-agnostic representations for effective feature extraction across training and testing tasks, the features from different domains still do not align under a unified distribution (as seen in Fig. 4). This misalignment limits the adaptability of FSDA-Former to testing tasks following FSL. To address this, we propose FSDA-TDA that combines pseudolabel learning with support optimization for DA.

First, we apply CVA to the test HSIs $\{X_t^1, X_t^2\}$ to produce a vector difference map M

$$M\left(i, j\right) = \sqrt{\sum_{B=1}^{b} \left(X_t^1\left(i, j, B\right) - X_t^2\left(i, j, B\right)\right)} \tag{10}$$

where (i, j) represents pixel coordinates. We then cluster M into three classes using K-means and randomly select m pixels from the smallest and largest clusters to represent the unchanged and changed classes, respectively, forming a pseudolabel set $\mathbf{U} = \{\mathbf{x_U}, \hat{\mathbf{y}}\}$. The few labeled samples from the test set are used as the support set, denoted as $\mathbf{S} = \{\mathbf{x_S}, \mathbf{y_S}\}$, and together with the pseudolabel set $\mathbf{U}$, they are treated as a new task ψ. After passing ψ through FSDA-Former, we obtain feature representations $\mathbf{z_S}$ for the support set and $\mathbf{z_U}$ for the pseudolabel set. Using $\mathbf{z_S}$, we compute class prototypes $\mathbf{A}$, then predict labels $\mathbf{y}'_\mathbf{U}$ for $\mathbf{U}$ by comparing the similarity of $\mathbf{z_U}$ to $\mathbf{A}$

$$\mathbf{y}'_\mathbf{U} = arg\max_c \frac{\mathbf{z}^{\mathbf{U}} \cdot \mathbf{A}}{||\mathbf{z}^{\mathbf{U}}|| ||\mathbf{A}||}. \tag{11}$$

We then compute cross-entropy loss L_U between predictions $\mathbf{y}'_\mathbf{U}$ and true pseudolabels $\hat{\mathbf{y}}$

$$L_U = -\sum \hat{\mathbf{y}} \log\left(\mathbf{y}'_\mathbf{U}\right). \tag{12}$$

After a few epochs of training, FSDA-Former begins to adapt to the test task domain. However, full adaptation to the test CD task is not yet achieved. The next step involves applying support optimization to further enhance DA and refine class prototypes for the test task.

We compute the feature $\mathbf{z}^{\mathbf{S}}$ for $\mathbf{S}$ of test task and then use (7) to calculate the class prototype $\mathbf{A}$. Using cosine similarity, we predict labels $\mathbf{y}'_\mathbf{S}$ for $\mathbf{z}^{\mathbf{S}}$

$$\mathbf{y}'_\mathbf{S} = arg\max_c \frac{\mathbf{z}^{\mathbf{S}} \cdot \mathbf{A}}{||\mathbf{z}^{\mathbf{S}}|| ||\mathbf{A}||}. \tag{13}$$

Finally, we compute the cross-entropy loss L_t between predictions $\mathbf{y}'_\mathbf{S}$ and true labels $\mathbf{y_S}$

$$L_t = -\sum \mathbf{y_S} \log\left(\mathbf{y}'_\mathbf{S}\right). \tag{14}$$

To prevent overfitting to the pseudolabels and support set of the test task, only a few epochs are used in both adaptation stages, during which the FSDA-MAE encoder is unfrozen to enhance generalization to the test task, completing DA and preparing the FSDA-Former for testing.

III. Experiments

A. Datasets

We performed experiments on six publicly accessible datasets: China,[1] USA[1] [4], river [50], Bay Area,[2] Santa Barbara,[2] and irrigated agriculture (IA) dataset.[3] These datasets

[1]https://rslab.ut.ac.ir/data

[2]https://citius.usc.es/investigacion/datasets/hyperspectral-change-detectiondataset

[3]https://github.com/SicongLiuRS/Hyperspectral-Change-Detection-Dataset-Irrigated-Agricultural-Area

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TABLE I
DETAILS OF FIVE HSIS-CD DATASETS

Dataset	Sensor	Imaged Times	Cover Land	Size ($h \times w \times b$)	Resolution	Main Change
China	EO-1	05. 2006 & 04. 2007	Yancheng, China	420×140×154	30m	land cover and river
river	EO-1	05. 2013 & 12. 2013	Jiangsu, China	431×241×198	30m	substance in river
USA	EO-1	05. 2004 & 05. 2007	Hermiston, USA	307×241×154	30m	land cover and river
Bay Area	AVIRIS	2013 & 2015	Bay Area, USA	600×500×224	20m	land cover
Santa Barbara	AVIRIS	2013 & 2014	Santa Barbara, USA	984×740 ×224	20m	land cover
IA	EO-1	05. 2004 & 05. 2007	Oregon, USA	180×225×159	30m	crops, soil

were captured by either the EO-1 Hyperion hyperspectral sensor or the AVIRIS (airborne visible/infrared imaging spectrometer). Table I provides detailed information about these datasets. These six datasets were used to pretrain the FSDA-MAE. We used the China, Bay Area, and Santa Barbara datasets to construct separate test tasks, evaluating the effectiveness of FSDA-Former. When testing on one dataset, the remaining five were used as auxiliary datasets for FSL. Each auxiliary dataset was randomly split into two training tasks, with $k = 5$ support samples and $|Q| = 20$ query samples per task. For the test task, $|Q|$ represents all pixels outside the support set.

B. Experimental Setup

The FSDA-MAE was pretrained using PyTorch 2.3.0 with CUDA 12.1 on an Intel[4] Core[5] i9-13900HX CPU and an NVIDIA GeForce RTX 4060 Laptop GPU under Ubuntu 22.04.4 LTS.

In generative pretraining for domain-agnostic representation, the FSDA-MAE used two Transformer encoder layers ($L = 2$) for both encoding and decoding. The patch size was $5 \times 5 \times 224$. The stride (s) was set to 5, with a spectral masking probability of 75%. The encoder's linear projection (LP) dimension d was set to 64, as was the LP dimension r in the first decoder layer. We trained the model with the AdamW optimizer (betas set to 0.9 and 0.95), over 4000 epochs with a learning rate of 0.00015. The encoder's hyperparameters remained constant across all training and testing phases.

For FSL and FSDA-TDA, we used the Adam optimizer with a learning rate of 0.0005. FSL was trained for 150 epochs, with the learning rate reduced by 0.1 every 20 epochs. Pseudolabel learning was conducted for five epochs, while support optimization was fixed at ten epochs across all datasets. For DA, 50 pseudolabels per class were employed.

Model performance in HSIs-CD was assessed using overall accuracy (OA), the kappa coefficient ($\mathcal{K}$), and the $F1$ score ($F1$), all derived from the confusion matrix values for true positive (TP), true negative (TN), false positive (FP), and false negative (FN). The OA is calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}}. \tag{15}$$

$\mathcal{K}$ measures performance considering class imbalance and is given by

$$\mathcal{K} = \frac{\mathrm{OA} - p_e}{1 - p_e} \tag{16}$$

$$p_e = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})}{(\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN})^2}. \tag{17}$$

The $F1$ score provides a balance between precision and recall, especially useful when dealing with imbalanced datasets, by

$$F1 = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{18}$$

$$\mathrm{Precision} = \frac{TP}{TP + FP} \tag{19}$$

$$\mathrm{Recall} = \frac{TP}{TP + FN}. \tag{20}$$

We evaluated the effectiveness of our proposed method by comparing it against seven benchmark models: CVA, SST-Former [30], HyGSTAN [25], DA-Former [32], HyperSIGMA [38], BiG-FSLF [45], CSIDC-FSL [46], CAFSL [47], and HGA2-FSL [51]. For a fair comparison, the supervised methods HyGSTAN and SST-Former were pretrained on 25 samples from each training task and then trained on five samples for the testing task.

C. Comparison With State-of-the-Art Methods

We compared the proposed FSDA-Former with several benchmark methods on the China, Bay Area, and Santa Barbara datasets. Because the source code of CAFSL [47] was not publicly available, we report the accuracy values given in its original paper. All other methods were run five times, and the average results are reported. The quantitative comparisons are listed in Table II, and the corresponding visual results are shown in Figs. 5–7.

On the China dataset, CVA produced the worst across all three metrics. Among the FSL methods, BiG-FSLF yielded the lowest OA and $\mathcal{K}$, followed by HGA2-FSL. CSIDC-FSL improved OA and $\mathcal{K}$ over HGA2-FSL by 0.88% and 3.8%, respectively, although its $F1$ was about 2% lower than those of BiG-FSLF and HGA-FSL. HyperSIGMA achieved performance comparable to CSIDC-FSL on all three metrics. DA-Former outperformed HyperSIGMA and achieved the highest $F1$. SST-Former and HyGSTAN, both based on transfer learning, showed similar results and surpassed DA-Former in OA and $\mathcal{K}$. The proposed FSDA-Former achieved the highest OA and $\mathcal{K}$, with the second-highest $F1$. In the visual results, CVA and DA-Former missed many changed regions, while HyGSTAN introduced substantial false alarms over large areas. DA-Former, HyperSIGMA, BiG-FSLF, and HGA2-FSL showed weak boundary preservation, resulting in blurred or inaccurate change outlines. By contrast, SST-Former, CSIDC-FSL, and FSDA-Former generated clearer change maps, with FSDA-Former showing fewer false detections.

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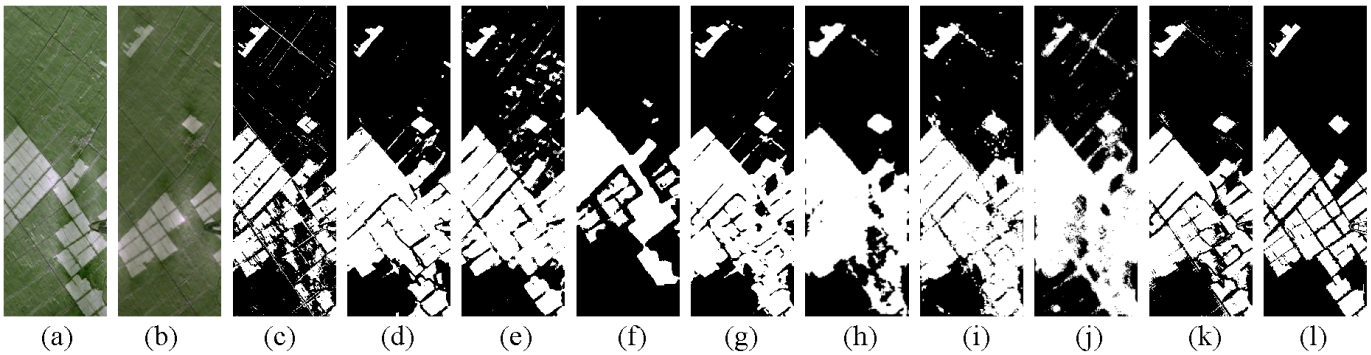


Fig. 5. HSIs-CD results on China dataset: (a) pseudocolor image of China data obtained in 2006, (b) pseudocolor image of China data obtained in 2007, (c) CVA, (d) SST-Former, (e) HyGSTAN, (f) DA-Former, (g) HyperSIGMA, (h) BiG-FSLF, (i) CSIDC-FSL, (j) HGA2-FSL, (k) FSDA-Former, and (l) groundtruth.

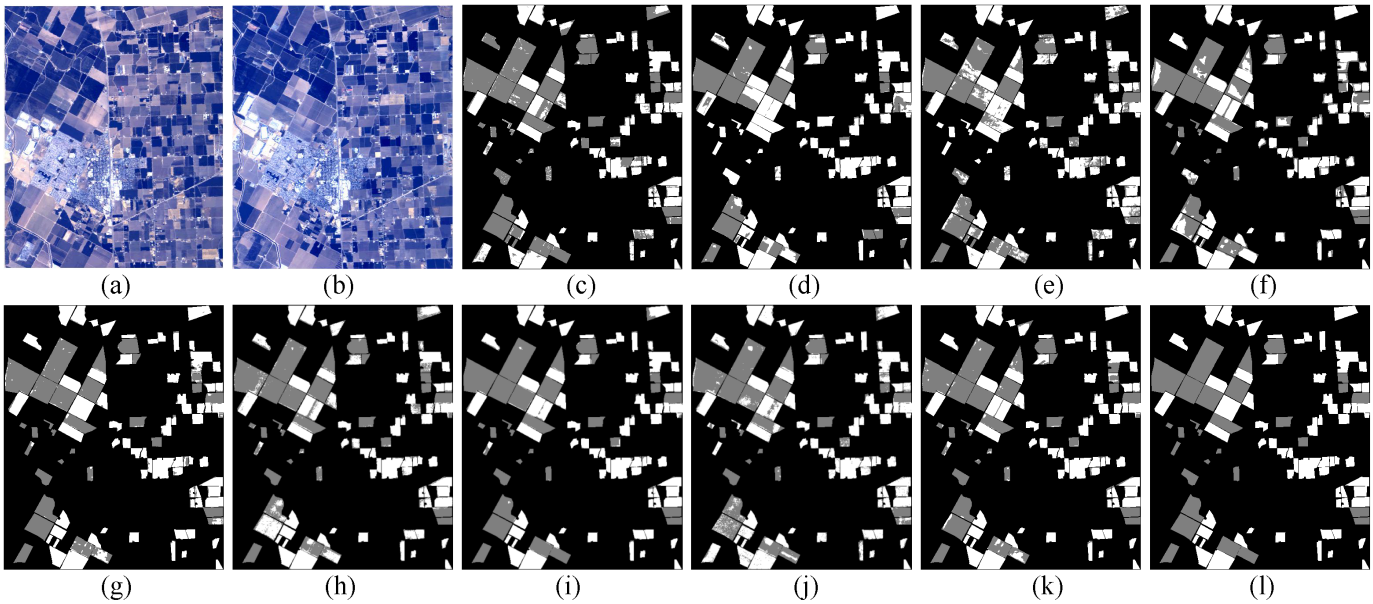


Fig. 6. HSIs-CD results on the Bay Area dataset: (a) pseudocolor image of Bay Area data obtained in 2013, (b) pseudocolor image of Bay Area data obtained in 2015, (c) CVA, (d) SST-Former, (e) HyGSTAN, (f) DA-Former, (g) HyperSIGMA, (h) BiG-FSLF, (i) CSIDC-FSL, (j) HGA2-FSL, (k) FSDA-Former, and (l) groundtruth.

TABLE II

COMPARISON OF OA, $\mathcal{K}$ AND $F1$ WITH STATE-OF-THE-ART METHODS ON THREE DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD. THE SYMBOL "-" INDICATES THAT THE RESULT IS NOT AVAILABLE

	China			Bay Area			Santa Barbara		
	OA	$\mathcal{K}$	F1	OA	$\mathcal{K}$	F1	OA	$\mathcal{K}$	F1
CVA	0.8749	0.6998	0.7882	0.8406	0.6844	0.8464	0.8325	0.6519	0.8598
SST-Former [30]	0.9045	0.7762	0.8881	0.8425	0.6792	0.8381	0.8437	0.6751	0.8375
HyGSTAN [25]	0.8933	0.7662	0.8954	0.8232	0.6462	0.8229	0.7895	0.5699	0.7843
DA-Former [32]	0.9009	0.7573	**0.9341**	0.8359	0.6702	0.8239	0.9366	0.8659	**0.9485**
HyperSIGMA [38]	0.8820	0.7453	0.8349	0.9412	0.8818	0.9398	0.9335	0.8712	0.9385
BiG-FSLF [45]	0.8750	0.7099	0.8549	0.9140	0.8280	0.9139	0.9149	0.8211	0.9105
CSIDC-FSL [46]	0.8868	0.7485	0.8334	0.9331	0.8661	0.9354	0.9312	0.8542	0.9098
CAFSL [47]	-	-	-	0.9249	0.8500	0.9267	0.9314	0.8558	0.9120
HGA2-FSL [51]	0.8780	0.7105	0.8588	0.8933	0.7837	0.8976	0.9007	0.7921	0.8901
FSDA-Former	**0.9019**	**0.8038**	0.9019	**0.9461**	**0.8923**	**0.9461**	**0.9404**	**0.8808**	0.9404

On the Bay Area dataset, SST-Former, HyGSTAN, and DA-Former performed poorly in terms of $\mathcal{K}$ and $F1$, even falling below the unsupervised CVA. On the contrary, the FSL methods, including HGA2-FSL, BiG-FSLF, CSIDC-FSL, and CAFSL, all achieved clear improvements over these methods. Among them, CSIDC-FSL delivered the best results, reaching 93.31% OA, 86.61% $\mathcal{K}$, and 93.54% $F1$. HyperSIGMA further improved upon CSIDC-FSL after few-shot fine-tuning. The

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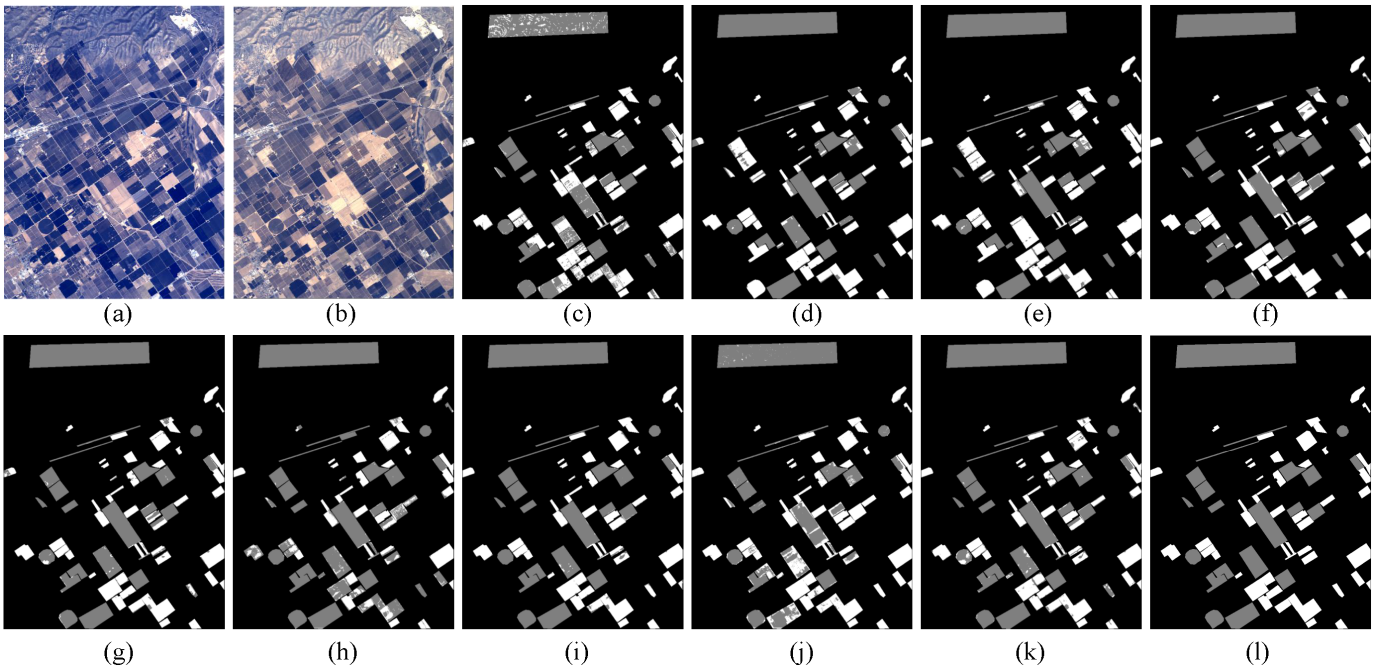


Fig. 7. HSIs-CD results on Santa Barbara dataset: (a) pseudocolor image of Santa Barbara data obtained in 2013, (b) pseudocolor image of Santa Barbara data obtained in 2014, (c) CVA, (d) SST-Former, (e) HyGSTAN, (f) DA-Former, (g) HyperSIGMA, (h) BiG-FSLF, (i) CSIDC-FSL, (j) HGA2-FSL, (k) FSDA-Former, and (l) groundtruth.

proposed FSDA-Former achieved the best performance across all three metrics. The visual results in Fig. 6 further support these findings. CVA, SST-Former, HyGSTAN, DA-Former, BiG-FSLF, and HGA2-FSL all suffered from large areas of missed or false detections. HyperSIGMA showed weak boundary delineation and was affected by noticeable salt-and-pepper noise. In comparison, CSIDC-FSL and FSDA-Former exhibited better detection results.

On the Santa Barbara dataset, SST-Former and HyGSTAN had the lowest accuracy, with CVA performing slightly better than both. Compared with SST-Former, HGA2-FSL improved OA, $\mathcal{K}$, and $F1$ by 5.7%, 11.7%, and 5.26%, respectively. BiG-FSLF further improved upon HGA2-FSL, while CSIDC-FSL and CAFSL achieved additional gains over BiG-FSLF. HyperSIGMA also improved $\mathcal{K}$ by about 2% over CSIDC-FSL. DA-Former obtained OA and $\mathcal{K}$ values close to those of HyperSIGMA while achieving the highest $F1$-score. FSDA-Former delivered the highest OA and $\mathcal{K}$, and its $F1$-score was only 0.81% lower than that of DA-Former. As shown in Fig. 7, CVA exhibited strong salt-and-pepper noise, whereas SST-Former, HyGSTAN, and HGA2-FSL produced large false detection regions. BiG-FSLF missed substantial changed areas. DA-Former and HyperSIGMA showed only small false detection regions. Among all methods, CSIDC-FSL and FSDA-Former produced the cleanest results, with minimal noise and error regions.

In conclusion, CVA, SST-Former, and HyGSTAN showed the weakest and least stable performance, with substantial noise and missed detections across datasets. DA-Former achieved competitive results on the China and Santa Barbara datasets, particularly in terms of $F1$-score, but performed poorly on the Bay Area dataset. HyperSiGMA also demonstrated its strengths with few-shot fine-tuning, showing solid performance. The few-shot methods BiG-FSLF, CSIDC-FSL, CAFSL, and HGA2-FSL also demonstrated good performance, but they did not perform as well as HyperSiGMA. Overall, FSDA-Former demonstrated the most balanced and reliable detection, making it the most effective in HSIs-CD tasks.

D. Ablation Studies

The ablation experiment in Table III provides insights into the contributions of FSDA-MAE, FSL, and FSDA-TDA on the overall performance.

1) *FSDA-MAE:* Across all datasets, FSL without FSDA-TDA benefited significantly from FSDA-MAE pretraining (① versus ④). Similarly, when FSDA-TDA was applied without FSL, FSDA-MAE still improved performance (② versus ⑥). The largest gains were observed when both FSL and FSDA-TDA were present, where FSDA-MAE led to the highest accuracy improvements (⑤ versus ⑦). In all cases, FSDA-MAE consistently enhanced model performance.
2) *FSL:* FSL consistently improved performance across datasets. Without FSDA-MAE, adding FSL on top of FSDA-TDA increased accuracy (② versus ⑤). With FSDA-MAE, FSL also delivered clear gains (③ versus ④). When both FSDA-MAE and FSDA-TDA were applied, FSL yielded the largest improvements (⑥ versus ⑦), confirming its contribution under all settings.
3) *FSDA-TDA:* FSDA-TDA further boosted performance across all datasets. Without FSDA-MAE, its inclusion improved FSL results (① versus ⑤). With FSDA-MAE but no FSL, FSDA-TDA also enhanced performance (③ versus ⑥). When combined with both FSL and FSDA-MAE, FSDA-TDA provided the highest accuracy gains (④ versus ⑦), confirming its effectiveness across configurations.

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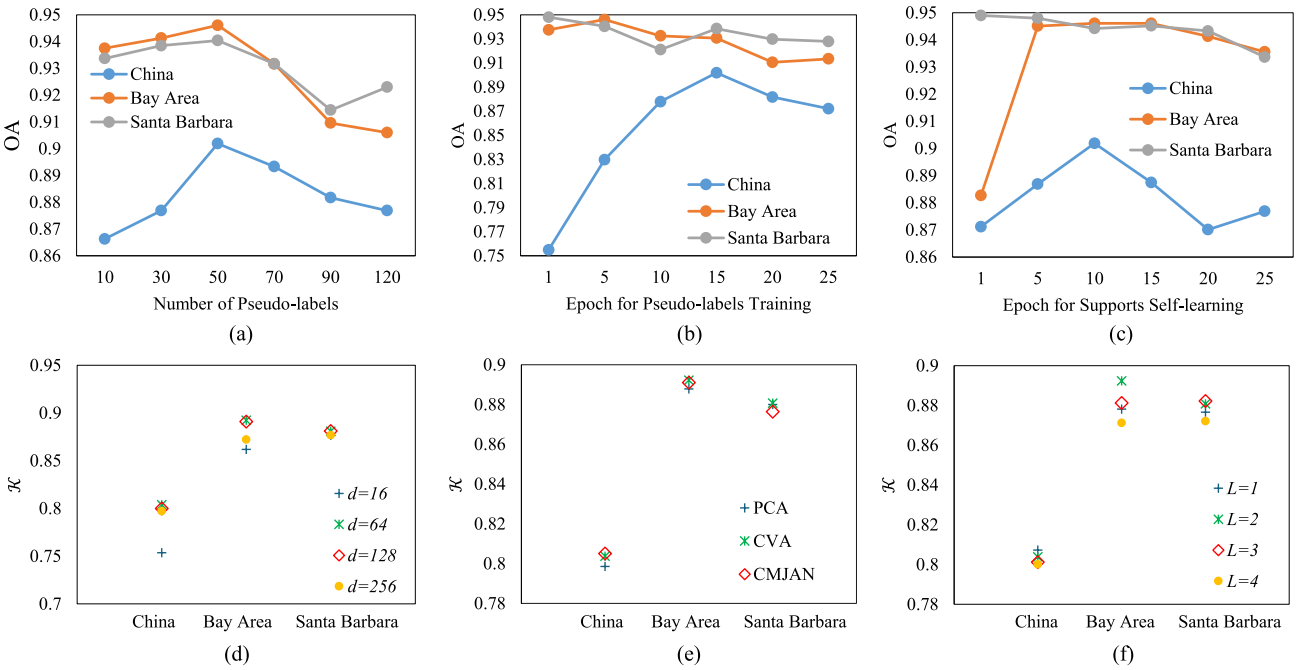


Fig. 8. Hyperparameter analysis of FSDA-Former. (a) Numbers of pseudolabels. (b) Epochs for pseudolabel learning. (c) Epochs for support optimization. (d) Different LP dimensions d. (e) Pseudolabel strategies. (f) Transformer encoder layers L.

E. Hyperparameters Studies

The hyperparameters of the FSDA-Former architecture were adopted directly from prior works [30], [38] and were not further optimized in this study. Instead, we focused on evaluating the impact of key training hyperparameters—the number of pseudolabels, pseudolabel learning epochs, and support optimization epochs—on the accuracy of FSDA-Former.

In pseudolabel learning, a larger number of pseudolabels increases both correct and incorrect samples, so balancing this is essential for optimal performance. To investigate this, we set the number of pseudolabels to [10, 30, 50, 70, 90, 120], with results shown in Fig. 8(a). Across all three datasets, OA initially increased as the number of pseudolabels rose, peaking at 50 before declining. Notably, in the initial growth phase, the OA for the China dataset showed a steeper increase compared to the Bay Area and Santa Barbara datasets, likely due to the greater domain shift present in the China dataset.

To prevent the FSDA-Former from falling into the pseudolabel trap during DA, we used a limited number of epochs for pseudolabel training. Analyzing the optimal epoch count is crucial. Here, epochs were set to [1, 5, 10, 15, 20, 25] to evaluate their effect, as shown in Fig. 8(b). In the China dataset, OA initially increased with the number of epochs before declining, while in the other two datasets, OA showed a general downward trend. This difference likely reflects the larger domain shift in the China dataset, which requires more adaptation steps, whereas the other two datasets benefit from fewer epochs due to smaller domain shifts. Overall, setting epochs to 15 proved most effective across datasets.

Support optimization with only five samples requires careful tuning of epochs to avoid overfitting with too many epochs or underfitting with too few. To evaluate this, we varied epochs across [1, 5, 10, 15, 20, 25], with results shown in Fig. 8(c). The experimental results reveal that for the China and Bay Area datasets, OA initially increases before declining, while for the Santa Barbara dataset, OA shows a consistent decline. Based on trends across all three datasets, we set this parameter to ten for optimal balance.

To investigate the effect of the LP dimension d, four commonly used settings, $d \in \{16, 64, 128, 256\}$, were evaluated. As illustrated in Fig. 8(d), $d = 64$ provides the best overall $\mathcal{K}$ on the three datasets. We can find that $d = 128$ also achieves a relatively high $\mathcal{K}$, but it introduces higher computational cost. Therefore, considering both detection performance and efficiency, $d = 64$ is adopted in FSDA-Former.

To analyze the effect of different pseudolabel strategies, we compared CVA+K-means, PCA+K-means, and CMJAN [52]. As shown in Fig. 8(e), the three methods yield similar $\mathcal{K}$ across the three datasets. We attribute this to the fact that pseudolabels are not directly used as ground-truth supervision for classifier training, but instead serve as an initialization signal for DA in FSDA-Former. Consequently, once the pseudolabels attain a reasonable level of reliability, the final adaptation performance becomes relatively insensitive to the pseudolabel strategy.

To investigate the effect of the number of Transformer encoder layers L, we conducted experiments with $L \in 1, 2, 3, 4$. The corresponding results are shown in Fig. 8(f). On the China dataset, $L = 1$ achieves the best $\mathcal{K}$, followed by $L = 2$. On the Bay Area dataset, the best $\mathcal{K}$ is obtained with $L = 2$. On the Santa Barbara dataset, $L = 2$ and $L = 3$ achieve the best $\mathcal{K}$. Overall, $L = 2$ yields the best performance across the three datasets.

F. Complexity Analysis

Another important factor in evaluating the practical applicability of HSIs-CD is computational complexity. To this end,

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TABLE III
ABLATION STUDY RESULTS FOR FSDA-FORMER, SHOWING THE EFFECT OF FSDA-MAE, FSL, AND FSDA-TDA ON OA AND $\mathcal{K}$ ACROSS THREE DATASETS. BEST RESULTS ARE HIGHLIGHTED IN BOLD

	FSDA-MAE	FSL	FSDA-TDA	China		Bay Area		Santa Barbara	
				OA	$\mathcal{K}$	OA	$\mathcal{K}$	OA	$\mathcal{K}$
①	✗	✓	✗	0.7269	0.4538	0.6490	0.2981	0.7538	0.5096
②	✗	✗	✓	0.6971	0.3942	0.6134	0.2269	0.6308	0.2615
③	✓	✗	✗	0.6981	0.3961	0.8827	0.7654	0.7394	0.4788
④	✓	✓	✗	0.8125	0.6250	0.8904	0.7807	0.7875	0.5750
⑤	✗	✓	✓	0.7692	0.5385	0.6856	0.3712	0.7615	0.5230
⑥	✓	✗	✓	0.8904	0.7808	0.9384	0.8769	0.9250	0.8500
⑦	✓	✓	✓	**0.9019**	**0.8038**	**0.9461**	**0.8923**	**0.9404**	**0.8808**

TABLE IV
COMPLEXITY ANALYSIS OF DIFFERENT METHODS ON CHINA DATASETS

Method	Params (K)	FLOPs (M)	Test time (s)
SST-Former	2498	145	7.52
HyGSTAN	354	9	5.15
DA-Former	398	30	10.46
HyperSIGMA	174596	29284	1317.75
BiG-FSLF	335	81	23.10
CSIDC-FSL	430	46	20.69
HGA2-FSL	240	188	185.66
FSDA-Former	126	28	11.13

we analyze the complexity of several deep learning-based methods, including SST-Former, HyGSTAN, DA-Former, HyperSIGMA, BiG-FSLF, CSIDC-FSL, HGA2-FSL, and FSDA-Former. Three metrics are used for comparison, namely the number of parameters (Params), floating-point operations (FLOPs), and test time. The corresponding results are presented in Table IV. In particular, the parameters and FLOPs are calculated based on a single patch from the China dataset, while the test time represents the total inference time for the entire China dataset.

As illustrated in Table IV, HyperSIGMA exhibits the highest computational cost, with 174 596K parameters, 29 284M FLOPs, and a test time of 1317.75 s, indicating that its strong representation capability is achieved at the expense of substantial computational overhead. On the contrary, HyGSTAN achieves the lowest FLOPs (9M) and the shortest test time (5.15 s), demonstrating high computational efficiency. However, its accuracy is relatively low under the few-shot setting. Compared with HyperSIGMA and the existing few-shot methods, including BiG-FSLF, CSIDC-FSL, and HGA2-FSL, FSDA-Former achieves the lowest parameters, FLOPs, and test time. In particular, it requires only 126K parameters, 28M FLOPs, and 11.13 s of test time. These results demonstrate that FSDA-Former is a lightweight and efficient framework for few-shot HSIs-CD, achieving competitive performance with low computational cost.

IV. CONCLUSION

In this study, we propose FSDA-Former, a model designed to improve few-shot HSIs-CD. Our approach demonstrates notable improvements across the China, Bay Area, and Santa Barbara datasets, particularly in addressing FSL problems under large distribution differences between training and testing tasks. The FSDA-Former architecture, combining FSDA-MAE initialization, FSL, and TDA, substantially improves accuracy, as confirmed by our ablation studies. Since generative pretraining of FSDA-MAE is time consuming, it is impractical to rerun pretraining on both the training data and a new test task. In future work, we will investigate an adapter-based transfer learning framework, in which the backbone pretrained on the training tasks is reused as a general feature extractor, while only lightweight adapter modules are optimized for the target task. This design is expected to effectively adapt the model to a new test task while reducing computational overhead and adaptation cost by avoiding full-model optimization.

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